# ML Offline Design

## Where We Stand

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Custom Bidding Agent · ML Lane · 2026

# What you'll know by the end

- What's already **decided**, and why
- What the offline design is actually **made of**
- What it needs **from each of you**
- What's still **open** — and what's blocking it

Two myths to drop today: **"it's just a model"** · **"we're blocked on everything"**

# Part 1

The system today

# Today vs. tomorrow

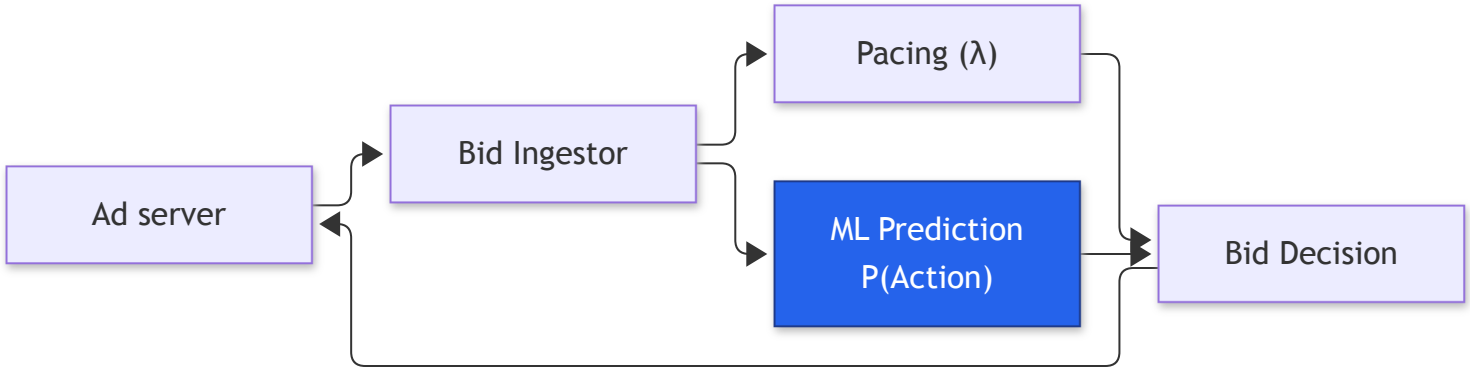
**Today** – every advertiser bids on awareness:

`Final Bid = λ × Max CPM`

**Tomorrow** – this project unlocks performance bidding:

`Final Bid = λ × Max CPA × P(Action)`

# Four services, one request




our lane: **ML Prediction**

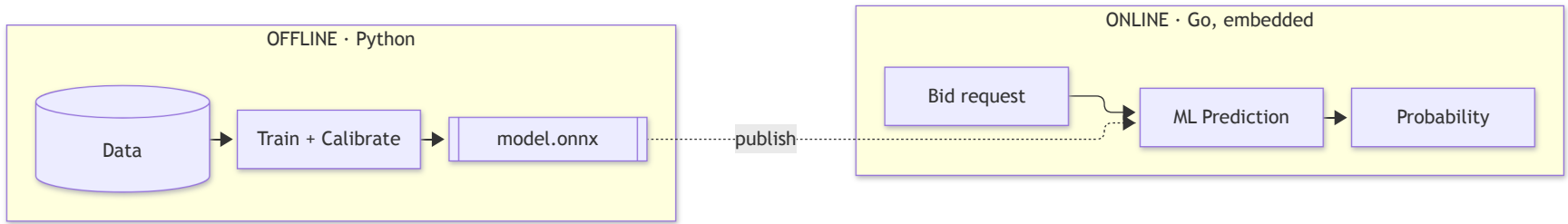
# Part 2

What’s already decided

# Four decisions, locked in

| Decision | Status |
|---|---|
| ML returns a **probability**, not a dollar value | Settled |
| Served as **ONNX**, embedded in the Go agent — no separate server | Settled |
| **Calibration is mandatory**, versioned with the model | Settled |
| Served artifact is **one self-contained file** (prep + model + calibration) | Settled |

# Two worlds, one bridge




ONNX is the only thing that crosses the bridge

# The latency budget

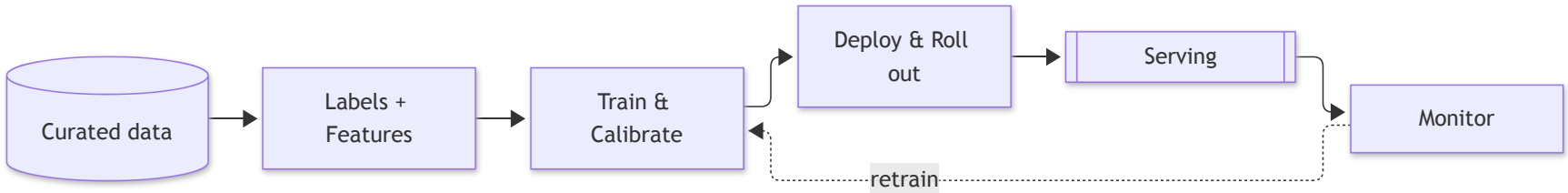
| Budget | Whose |
| --- | --- |
| **10 ms**, end-to-end HTTP | the ad server's hard timeout — whole agent |
| **~3 ms**, p99 | the ML step's own slice — our firm target |

a budget inside a budget — ingestor, pacing, and decision share the rest

# Part 3

What the offline design is made of

# The whole pipeline, end to end



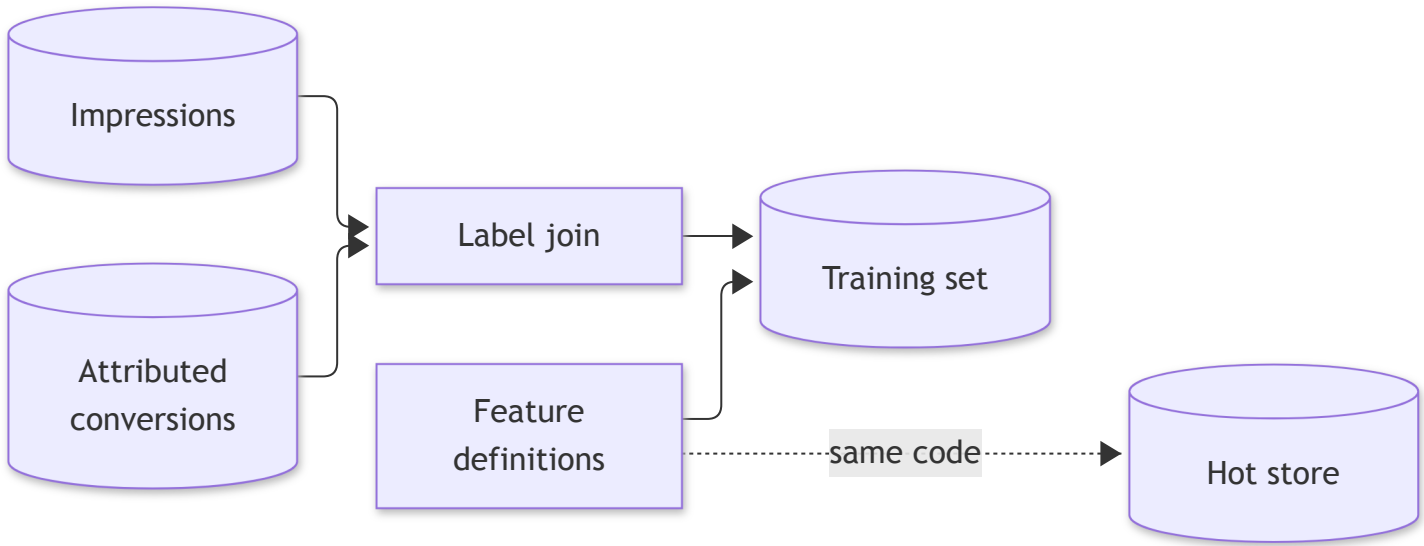

five stages – each one gets a slide, briefly

# Data in

- Impression-level training data
- Attribution labels (conversions)
- Aggregate performance data
- Delivery-control feed
- Bidder-response data

what we **require** from these is fixed; who **produces** them is still open

# Building a training set

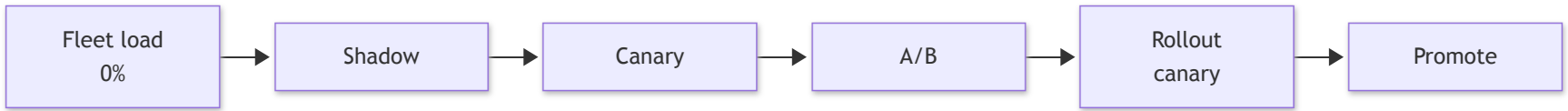



one feature definition feeds both training and serving – no drift

# Train & calibrate

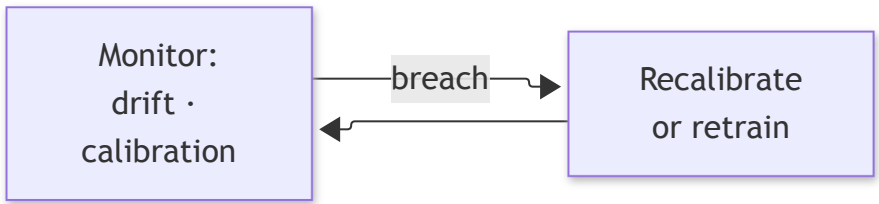
- Gradient-boosted trees, exported to **ONNX**
- Calibration **baked in**, versioned with the model as one pair
- Gate before anything ships: ranking must not regress, calibration must clear an absolute bound

# Deploy & roll out




2 human approvals on the way up · automatic abort anywhere on a guardrail breach

# Monitor & retrain

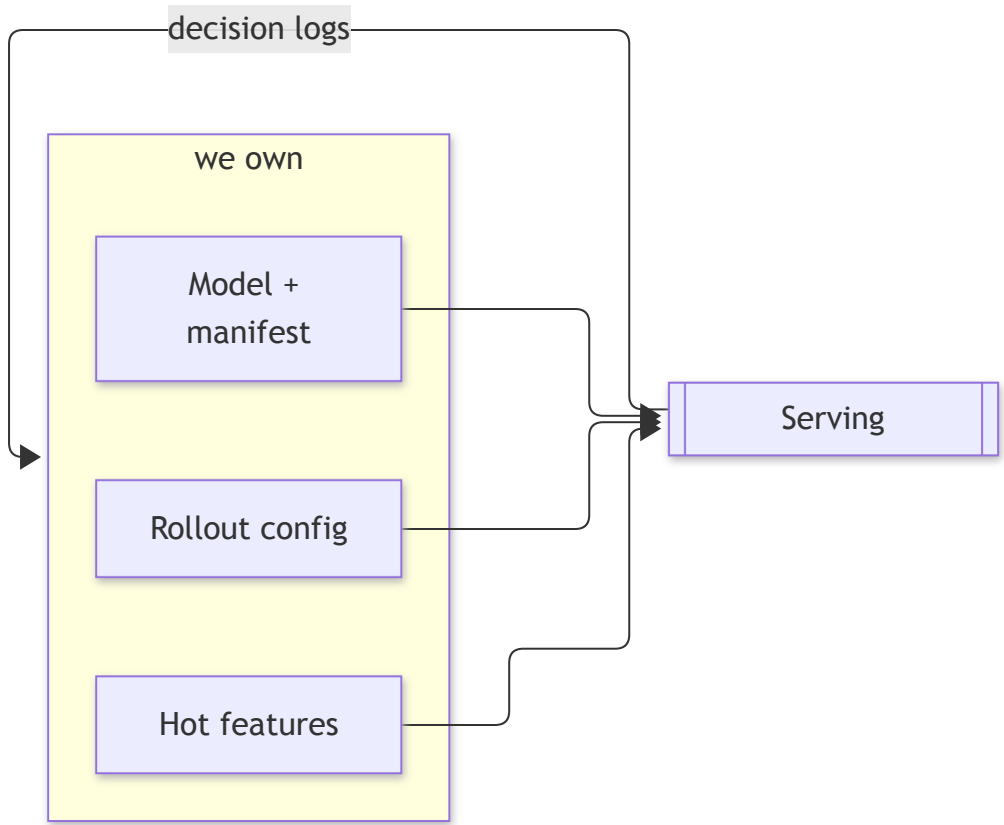



- Calibration breach → cheap **recalibrate-only** refresh
- Drift breach → **full retrain**

# Part 4

The contract with serving

# Four touchpoints



# Who writes, who reads

| Touchpoint | We write | Serving reads | Changes on |
|---|---|---|---|
| Artifact + manifest | ✓ | ✓ | new model version |
| Rollout config | ✓ | ✓ | new rollout state |
| Hot-feature store | ✓ | ✓ | every 5–10 min |
| Decision logs | ✓ (ingest) | ✓ (emits) | every scored candidate |

# Part 5

Infra this needs

# The stack

| Layer | Status |
| --- | --- |
| Cloud platform | confirmed |
| ML compute, tracking, model registry | confirmed |
| Specific tooling choices within it | ours to pick |
| Our own orchestration instance | ours; reuse-vs-new still open |
| Fast key-value store for hot features | needed; exact service deferred |

# New things to stand up

- A **pipeline-framework repo** — one folder per model pipeline
- A **versioned workload package** — training, calibration, ETL, feature code
- A **fast store** reachable by both our jobs and the serving agent
- No standing compute — pipelines submitted at trigger time only

# Part 6

What’s still open

# The two real blockers

1. **Attribution window + method** — what exactly we're training on
2. **Cross-boundary data-access mechanism** — tables are named, how they move isn't

both need an answer from the client

# Everything else

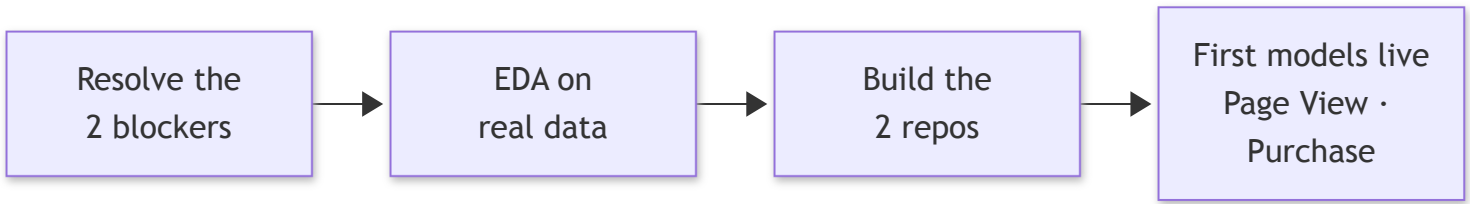
| Category | Roughly |
|---|---|
| Needs the client | ~13 items |
| Settles once EDA runs on real data | ~10 items |
| Ours to decide, no dependency | ~7 items |
| Asks on the serving side | ~8 items |

full list lives in the design register, not here

# Part 7

What happens next

# The path forward




no fixed dates yet – still discovery phase

# Takeaways

- The design is **complete and concrete** — not a sketch
- **Two client answers** unlock EDA and the build
- The contract with serving is **fixed** — both sides build in parallel

**Thank you — questions?**